

Assessing Data Completeness in OpenStreetMap using the Geometrical and Temporal Properties of Mapping Patterns

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Since the launch of OpenStreetMap (OSM), the quality of its data has been a prominent study subject. Among the various aspects of data quality, data completeness – i.e., whether all real-world geographic features in a given extent exist in the dataset – remains a central challenge [1]. Existing approaches to assessing the completeness of data in OSM can be broadly grouped into three main groups: extrinsic comparison against a reference dataset [2-5]; intrinsic analysis of some proxy variable in the data [6-8]; and statistical methods [9-12].

Extrinsic analysis, though the best estimator, requires an authoritative dataset, thus limiting this approach's applicability in regions where official or proprietary data are unavailable, outdated, or costly. Intrinsic methods compare OSM's edit history (e.g. edit frequency) with human development proxies (such as GDP) and allow for better scalability and flexibility between regions but tend to reflect local activity more than mapping outcomes. Statistical modeling approaches such as fitting Sigmoidal curves to cumulative road lengths [11] and applying Deep Learning Algorithms to detect missing features [12] require good training data and may not translate well over regions of different characteristics.

This study presents a novel intrinsic completeness metric based on the evolution of mean feature geometry, that requires only aggregate feature statistics and no external source. The method uses cumulative time series data, but differs from prior studies, e.g. [7, 11], by focusing on the relative change in the average value instead of the absolute cumulative value, accounting for both feature additions and geometry alterations. The underlying assumption draws on the works by Corcoran et al. [13] and Strano et al. [14], showing that road networks are typically mapped in a process in which exploration is inherited by densification. In other words, contributors initially add the largest, most prominent features (major roads, large and central buildings) and gradually move on to smaller and more detailed elements (minor roads, small structures). Thus, we can expect the average feature size (area or length) to decrease over time, until it converges to some general threshold such that the data can be classified "complete" (or "complete enough" for usage). Simply put, we expect to see the temporal signature of the average feature size (denoted $\bar{S}_t$) stabilize over time. For universal

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applicability and comparative analysis between regions, the metric is normalized by maximum absolute scaling, i.e. the largest average addition, in each time series (denoted $\overline{NS}_t$), which fits similarity-based methods using Euclidian Distance [15].

In our approach, a geographical region is classified "complete" if metric values present a stable saturation period T , identified via three conditions: $\overline{NS}_t < \alpha \ \forall t \in T$ where α is the convergence threshold; $|T| \geq \beta$ where β is the minimal duration for T ; and $C_T < \gamma$, where γ is the maximal threshold for the relative number of features added in T ($C_T = \frac{C_{\max(t)}}{C_{t_0 \in T}}$, C_t being the number of features in time t). In addition, a fourth condition is applied to prevent false negatives when external disruptions (e.g. a data export or HOT campaign) alter the signal: if $\frac{\min(\Delta C_t)}{\max(\Delta C_t)} \leq \rho : t \in T$, the saturation period is re-defined starting in $\max(\Delta C_t)$, and revalidated via β and γ . Using these validation tests, we can define whether a region has had minimal changes in the average feature size along with minimal feature additions for a sufficient time, suggesting it has been mostly mapped.

As an exploratory investigation of the approach's sensitivity to large-scale data events, we implemented the method for a sample of 446 regions categorized by distinct mapping dynamics [16]. Applying the method to both road (way features, *highway=* tag*) and building (way features, *building=* tag*) datasets suggested that the method is robust to all mapping dynamics tested.

The main experiment reported here is for a buildings sample of 266 urban and rural localities in Israel, comparing the results against the Survey of Israel (SOI) official building dataset [17] to test the method's performance against authoritative data. For this, we compute in each locality the ratio and difference between OSM final aggregated statistics (obtained using the OHSOME API [18]) and the equivalent SOI statistics, namely feature count, cumulative feature area and average feature area ($C_{\max(t)}$, $S_{\max(t)}$ and $\bar{S}_{\max(t)}$ respectively). Our implementation sets $\alpha = \rho = 0.1$, $\beta = 2$ and $\gamma = 1.5$. All plots and analysis steps are available at [19] for reproduction and verification.

The results show that for localities labeled as complete, both cumulative area and count ratios present highly accurate distributions, with the highest density around 90 – 100% similarity and less than 10% of the samples with under 40% similarity. However, localities labeled incomplete present a more bimodal shape, with ~50% of localities having at least 80% similarity between datasets, indicating a large proportion of false negatives (Fig. 1a,b). A possible explanation is miscalibration of the α parameter, as backed by the failed saturation condition: more than 90% ($n = 179$) of all localities had an insufficient stable period ($|T| \geq \beta$), with the majority having $|T| = 0$, suggesting a small $\bar{S}_t$ was used for normalization, resulting in high normalized values which did not converge. Furthermore, both metric comparison distributions ($\bar{S}_{\max(t)}$ ratio and difference) show a right-skewed shape with the highest densities around 100% and 0 (ratio and difference, respectively), backing the exploration-densification assumption. This suggests a viable, albeit restrictive metric to classify complete regions, which requires adjusted calibration.

Further inspection of results show that localities labeled as complete are predominantly cities (53% of all complete localities, which account for 47% of all cities), while local and regional councils show substantially lower but comparable rates (16% and 23% from their type, respectively; Figure 1c). Additionally, complete localities tend to be smaller than incomplete ones, with the latter exhibiting a more pronounced right-skew.

Geographically, the more rural North and Haifa regions contain a higher proportion of localities classified as incomplete, whereas the densely populated Gush Dan and Center regions exhibit a more balanced distribution (Figure 1d). This suggests the method to perform better in smaller, densely built localities rather than larger rural areas, presenting another possible explanation for the large proportion of false negatives.

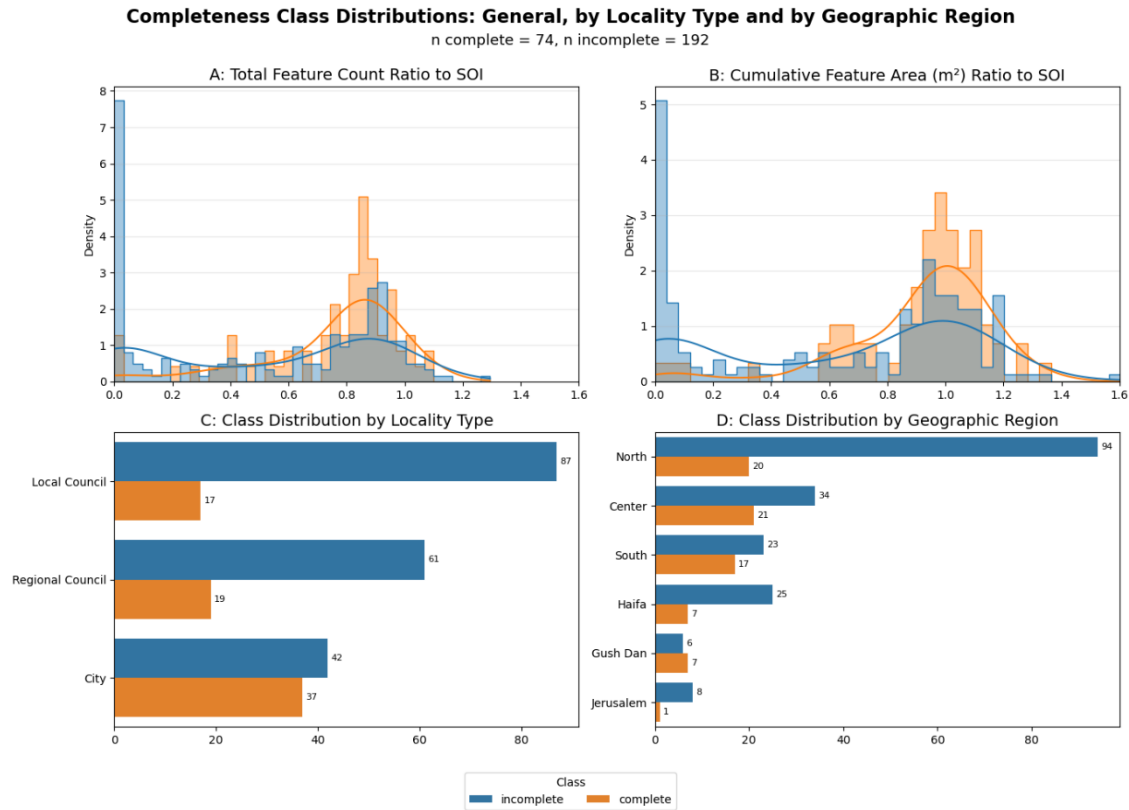


Figure 1. Completeness class distributions on the Israel locality datasets. Figures A and B present a normalized density histogram with KDE lines of the count and area ratios ($C_{\max(t)}$, $S_{\max(t)}$), and figures C and D present classification proportions in different locality types and regions.

The study contributes to OSM Science in several ways. First, it introduces the geometric properties of mapped features — specifically, their average size — as a completeness signal, a dimension absent from existing literature. The method can be implemented for various "way" feature types and applied for different confidence intervals (by calibrating all parameters), allowing a robust and flexible completeness assessment. Furthermore, the method may be adapted to assess the temporal quality of the data — perhaps by analyzing the curve's derivative or by defining different validation tests to locate benchmarks along the temporal series. Second, the metric's data requirements are exceptionally lightweight: it needs only aggregate statistics at regular time intervals. This makes it computationally inexpensive and universally applicable in principle, enabling both practitioners and the research community to define the reliability of an analyzed extent. However, as different regions present a variety of different geometries, calibration of parameters requires further exploration upon different regions in the world. Thus, future work should investigate parameter calibration and additional methods to reduce false negatives in rural areas.

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